Integer Models

EES 4891-06/5891-01
Bayesian Statistical Methods
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Class #16: Monday March 16, 2026

Integer Models

Counting Events

- Counting events:
  - Outcome variable is a whole number (0, 1, 2, ...)
  - Definition: **treatment condition** is the combination of all the predictor variables.
    - Each combination is a different treatment condition.
  - Two approaches:
    - Treat each event separately, with a probability p that depends on the *treatment condition* (predictor variables).
      - Outcome is 0 or 1
        - **Bernoulli distribution** Like a coin toss with probability p of getting heads
        - Equivalent: **binomial distribution** with one trial $\text{Binomial}(1, p)$
    - Group events by *treatment condition* and sum the events
      - Outcome is 0, 1, 2, ...
      - **Binomial distribution** $\text{Binomial}(N, p)$, where N is the number of trials for the treatment condition.

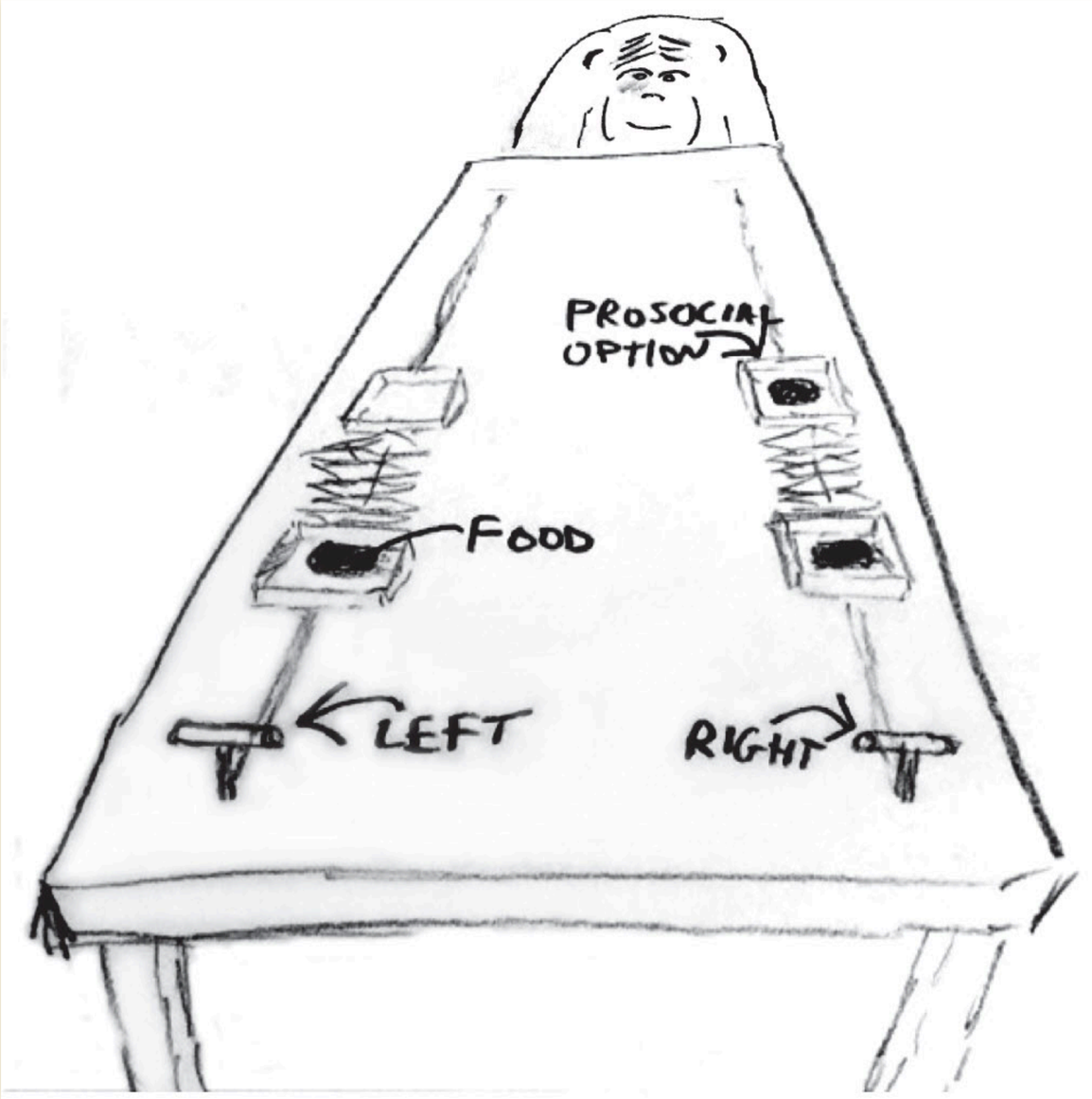
Analyzing Counting Events

- The quadratic approximation `quap()` only works well for *Normal* posterior distributions.
- *Bernoulli*, *binomial*, and *Poisson* distributions are very different to *Normal* distributions, so `quap` doesn't work.
- Use MCMC analysis (`ulam`).
- Homework exercise 11M7 examines the difference between `quap` and `ulam` for counting data.

Example: Chimpanzee Altruism

Chimpanzee Altruism

- A chimpanzee pulls either the right or left lever to get food
  - One lever is *pro-social*: When the chimp gets the food, a separate tray of food is sent to the other side of the table
  - Treatment conditions:
    1. Pro-social lever on the left or right
    2. The other side of the table is empty, or has a *partner* chimpanzee
  - Count the number of times the chimp pulls the *pro-social* lever.



Data

```
library(rethinking)
data(chimpanzees)
d <- chimpanzees
```

- `actor`: Which chimpanzee is pulling the levers (1–7)
- `condition`: Is there a partner (0 = no, 1 = yes)
- `prosoc_left`: Which side is the *pro-social* lever on? (0 = right, 1 = left)
- `pulled_left`: Outcome variable. Did the chimp pull the left lever (0 = no, 1 = yes)
- Create a `treatment` variable:
  1. pro-social on the right, no partner
  2. pro-social on the left, no partner
  3. pro-social on the right, partner
  4. pro-social on the left, partner

```
d <- d |> mutate(treatment = 1 + prosoc_left + 2 * condition)
```

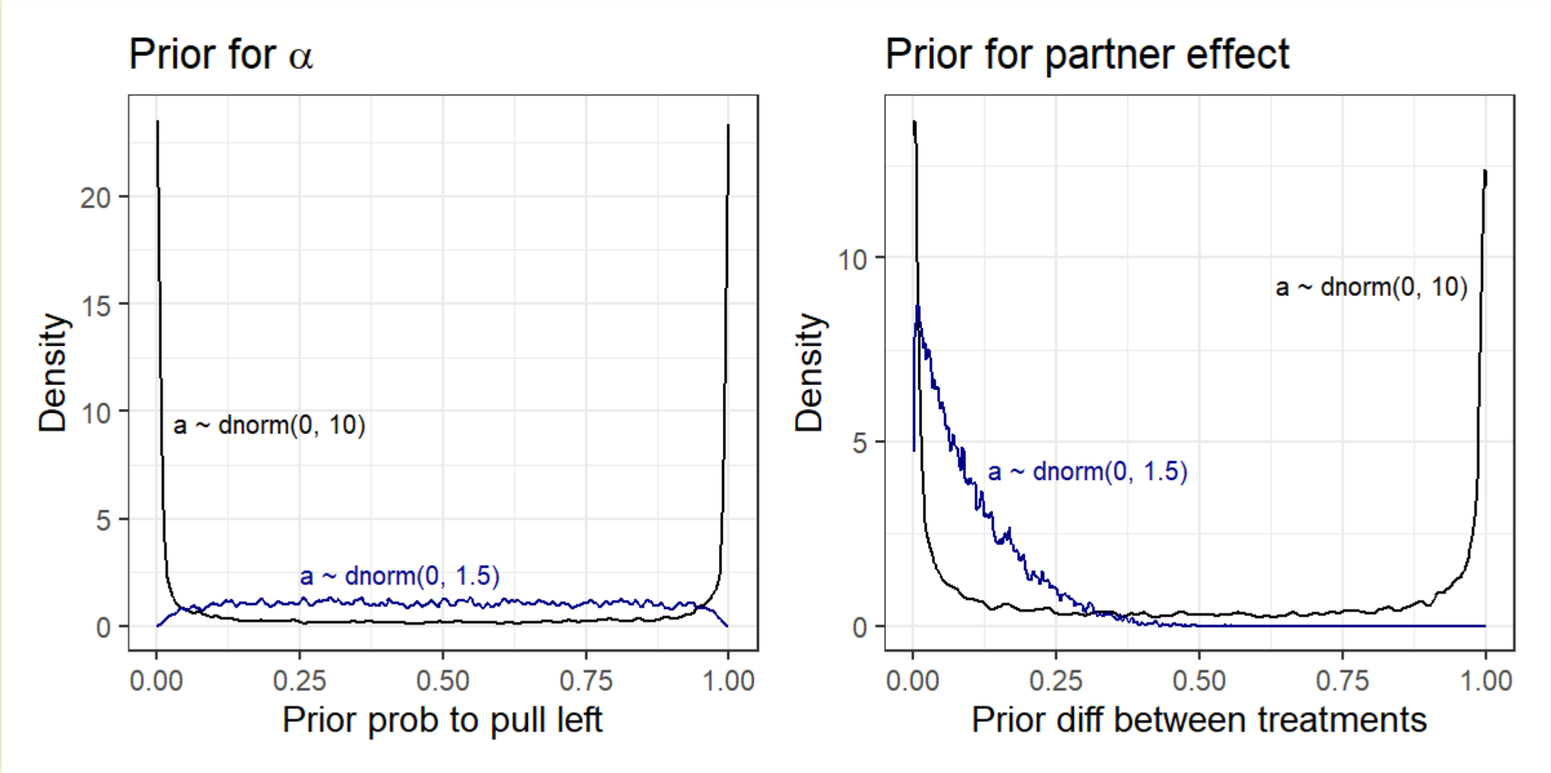
Model

$$
\begin{aligned}
L &\sim \text{Binomial}(1, p) \\
\text{logit}(p) &= \alpha_{\text{actor}} + \beta_{\text{treatment}} \\
\alpha &\sim \text{Normal}(?, ?) \\
\beta &\sim \text{Normal}(?, ?)
\end{aligned}
$$

- Each chimp has a different α, but the same $\beta_{\text{treatment}}$.
  - In Chapter 13 (Multilevel models) we'll make a model with different β for each chimp.

Choosing priors

- Wide (uninformative) vs. narrow (informative) priors:



Fit the model

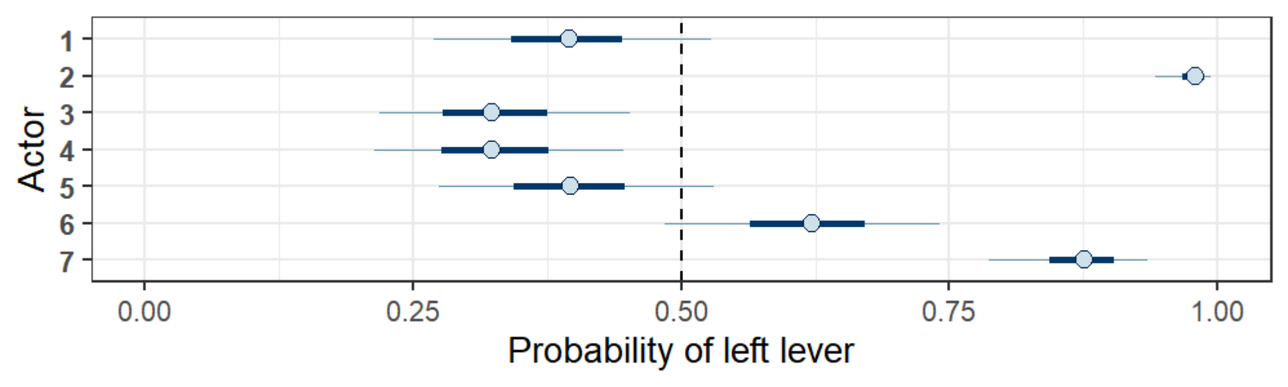
```
dat_trimmed <- d |> select(pulled_left, actor, treatment) |>
  mutate(treatment = as.integer(treatment))

mdl_chimp <- ulam(
  alist(
    pulled_left ~ dbinom(1, p),
    logit(p) <- a[actor] + b[treatment],
    a[actor] ~ dnorm(0, 1.5),
    b[treatment] ~ dnorm(0, 0.5)
  ), data = dat_trimmed, chains = 4, cores = 4, log_lik = TRUE)
```

```
precis_show(precis(mdl_chimp, pars = c("a", "b"), depth = 2,
                   digits = 2))
```

```
##       mean   sd  5.5% 94.5% rhat ess_bulk
## a[1] -0.43 0.33 -0.98  0.10 1.00   744.24
## a[2]  3.91 0.74  2.80  5.14 1.00  1237.37
## a[3] -0.73 0.33 -1.26 -0.21 1.00   798.28
## a[4] -0.74 0.33 -1.29 -0.23 1.00   734.43
## a[5] -0.43 0.33 -0.95  0.11 1.00   621.61
## a[6]  0.49 0.34 -0.05  1.03 1.00   818.88
## a[7]  1.97 0.41  1.32  2.65 1.00  1044.78
## b[1] -0.06 0.30 -0.52  0.43 1.00   716.99
## b[2]  0.47 0.28  0.01  0.93 1.00   710.40
## b[3] -0.40 0.29 -0.84  0.08 1.00   709.61
## b[4]  0.36 0.29 -0.10  0.81 1.01   661.86
```

What does this mean?

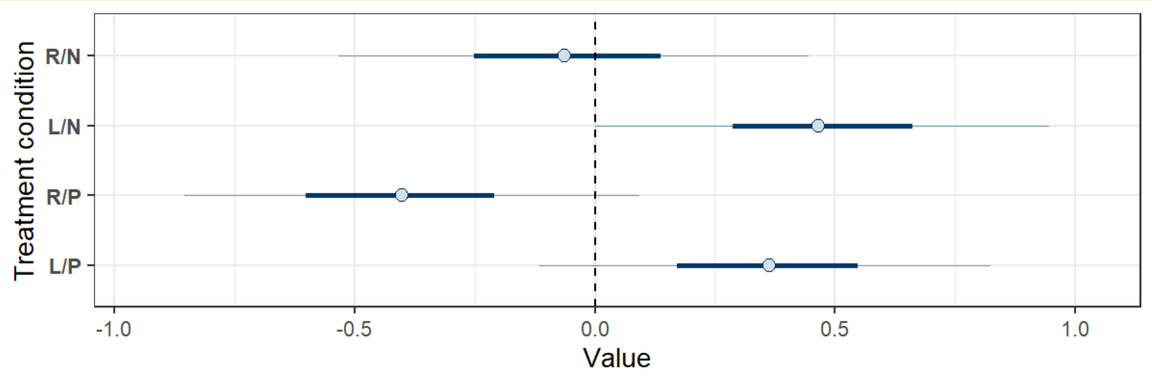
- α parameters:



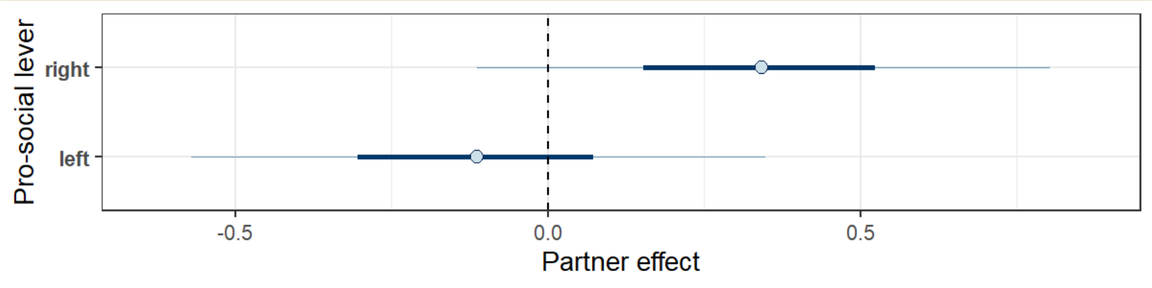
- This shows the posteriors with 50% and 95% highest-density intervals averaged over all treatment conditions
- What does it mean?
  - 3 chimps are more likely to pull the left lever
  - 4 chimps are more likely to pull the right lever
  - This may reflect handedness, especially for chimp #2.

Partner Effects

- Label treatment effects and compare the β parameters
  - Which lever is pro-social? / Is there a partner?



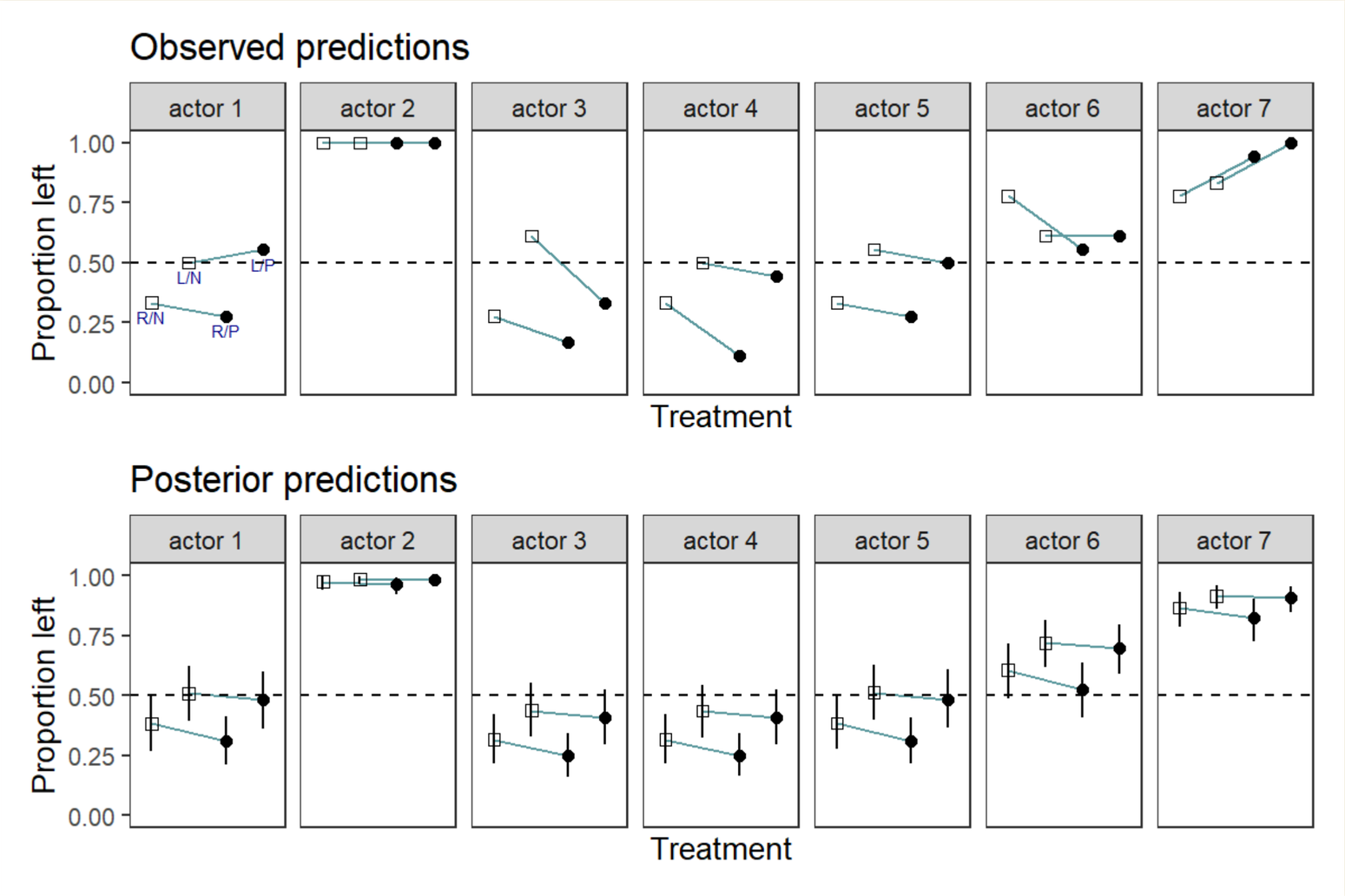
- Difference between partner and no-partner




```
library(tidybayes)
library(tidybayes.rethinking)
library(bayesplot)

treatment_levels = c("R/N", "L/N", "R/P", "L/P")

post <- spread_draws(mdl_chimp, a[actor], b[treatment]) |>
  mutate(treatment = treatment_levels[treatment]) |>
  select(-a) |>
  pivot_wider(names_from = "treatment", values_from = "b")

mcmc_intervals(post, pars = vars(contains("/"))) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(x = "Value", y = "Treatment condition")
```

- Notice the opposite sign for right and left
- β is the change in log-odds to pull the left lever:
  - Pro-social for the “L” treatment, anti for “R”.

Posterior Predictions



- A partner does not consistently change the probability of pulling the pro-social lever
- The model does not accurately predict the effect of adding a partner.
- The biggest effect seems to be the intercept, followed by the pro-social lever
- The intercept effect suggests that handedness matters more than pro-social motives

Aggregated Counts

Aggregated Chimp Model

- Aggregate data

```
d_agg <- d |> group_by(actor, treatment) |>
  summarize(left_pulls = sum(pulled_left), trials = n(),
            .groups = "drop")
```

- Now fit a model to it

```
mdl_chimp_agg <- ulam(
  alist(
    left_pulls ~ dbinom(trials, p),
    logit(p) <- a[actor] + b[treatment],
    a[actor] ~ dnorm(0, 1.5),
    b[treatment] ~ dnorm(0, 0.5)
  ), data = d_agg, chains = 4, cores = 4, log_lik = TRUE)
```

Comparing the models

- Compare the models

```
compare(mdl_chimp, mdl_chimp_agg, func = PSIS)
```

```
## Warning in compare(mdl_chimp, mdl_chimp_agg, func =
PSIS): Different numbers of observations found for at least
two models.
## Model comparison is valid only for models fit to exactly
the same observations.
## Number of observations for each model:
## mdl_chimp 504
## mdl_chimp_agg 28
```

```
## Some Pareto k values are high (>0.5). Set pointwise=TRUE
to inspect individual points.
```

```
##                   PSIS        SE    dPSIS      dSE   pPSIS
weight
## mdl_chimp_agg 114.0303  8.477327   0.0000       NA 8.33752
1.000000e+00
## mdl_chimp     532.3466 18.937491 418.3164 40.54771 8.56172 1.457995e-
91
```

- Why the big difference?
  - Imagine you toss a coin 9 times and get heads the first 6 times and then 3 tails:

$$P(1,1,1,1,1,0,0,0) = p^6(1-p)^{9-6}$$

  - But the binomial probability $P_{\text{binomial}}(6|9,p)$ considers all the ways you could get 6 heads in 9 tosses:

$$P(6|9,p) = \frac{6!}{6!(9-6)!}p^6(1-p)^{9-6}$$

    - This factor is the reason we can't easily compare a Bernoulli model of the individual events with a binomial model of the aggregated counts.

Graduate School Admissions

Graaduate School Admissions

- Famous study: P.J. Bickel, E.A. Hammel, & J.W. O'Connell, "Sex Bias in Graduate Admissions: Data from Berkeley", *Science* **187**, 398–404 (1975), doi: 10.1126/science.187.4175.398
- Research Question: Is there evidence of gender bias in graduate-school admissions?
- The paper examined applications to graduate school at the University of California Berkeley for the Fall 1973 admissions cycle.
  - 12,763 applications to 101 graduate programs
  - This data represents a subset of 4526 applications to 6 departments.
    - Men: 1198 of 2691 applicants were admitted (44.5%)
    - Women: 557 of 1835 applicants were admitted (30.4%)
    - But men and women tended to apply to different departments...

Data Analysis

```
data(UCBadmit)
d_grad <- UCBadmit |>
  mutate(applicant.gender =
           ordered(applicant.gender,
                   levels = c("male",
        "female")),
         gid = as.integer(applicant.gender))
```

```
rename(d_grad, gender = applicant.gender) |>
  summarize(admit = sum(admit),
            reject = sum(reject),
            applications = sum(applications),
            ratio = admit / applications,
            .by = "gender") |>
  kable(digits = 2)
```

gender	admit	reject	applications	ratio
male	1198	1493	2691	0.45
female	557	1278	1835	0.30

```
kable(d_grad)
```

dept	applicant.gender	admit	reject	applications	g
A	male	512	313	825	
A	female	89	19	108	
B	male	353	207	560	
B	female	17	8	25	
C	male	120	205	325	
C	female	202	391	593	
D	male	138	279	417	
D	female	131	244	375	
E	male	53	138	191	
E	female	94	299	393	
F	male	22	351	373	
F	female	24	317	341	

Model the Data

```
dat_list <- select(d_grad, admit, applications, gid)
```

```
mdl_gs <- ulam(
  alist(
    admit ~ dbinom(applications, p),
    logit(p) <- a[gid],
    a[gid] ~ dnorm(0, 1.5)
  ), data = dat_list, chains = 4, cores = 4)
```

```
precis_show(precis(mdl_gs, depth = 2, digits = 2))
```

```
##       mean   sd  5.5% 94.5% rhat ess_bulk
## a[1] -0.22 0.04 -0.28 -0.16 1.00  1391.29
## a[2] -0.83 0.05 -0.91 -0.75 1.01  1513.83
```

- Gender ID:
  - 1: Men
  - 2: Women

Interpreting the significance

```
precis_show(precis(mdl_gs, depth = 2, digits = 2))
```

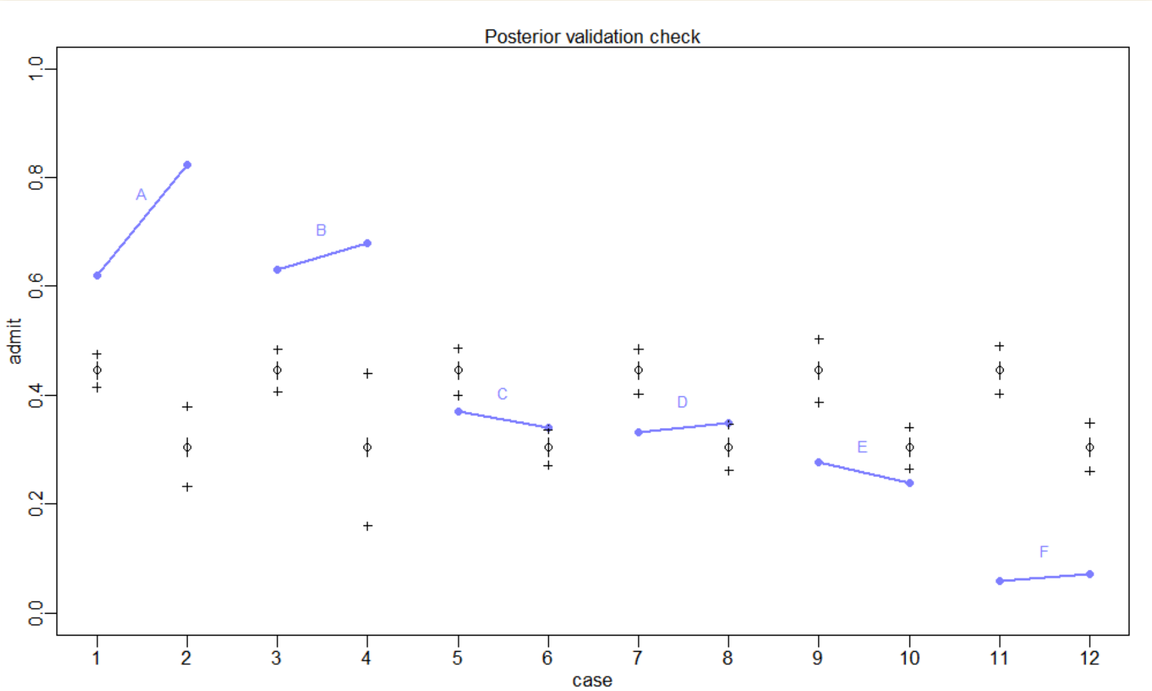
```
##       mean   sd  5.5% 94.5% rhat ess_bulk
## a[1] -0.22 0.04 -0.28 -0.16 1.00  1391.29
## a[2] -0.83 0.05 -0.91 -0.75 1.01  1513.83
```

- Gender ID:
  - 1: Men
  - 2: Women
- What is the significance of the gender difference in a?
  - Relative difference on logit scale
  - Absolute difference in odds of getting admitted

```
post <- tidy_draws(mdl_gs)
diff_a <- pull(post, "a[1]") - pull(post, "a[2]")
diff_p <- inv_logit(pull(post, "a[1]")) - inv_logit(pull(post, "a[2]"))
precis(list(diff_a = diff_a, diff_p = diff_p))
```

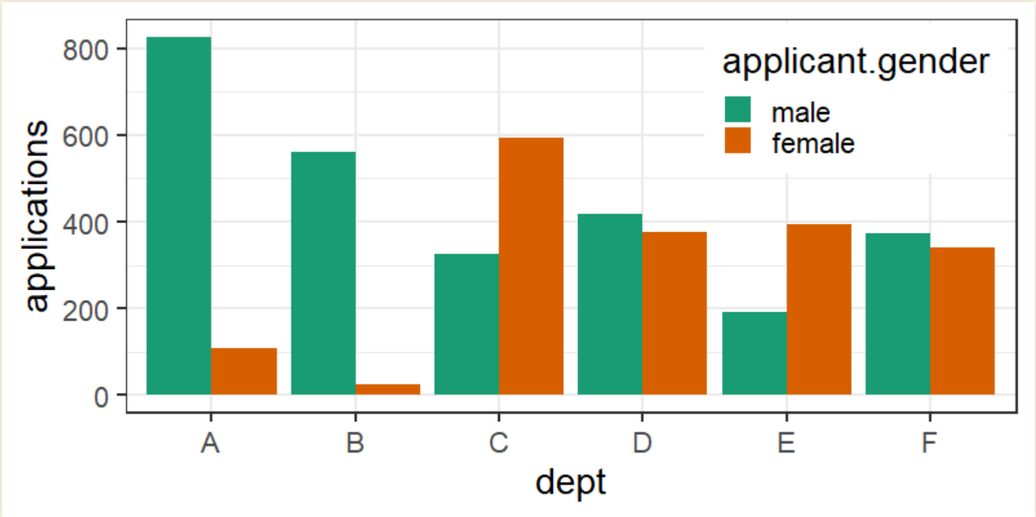
```
##             mean         sd      5.5%     94.5%   histogram
## diff_a 0.6100870 0.06204708 0.5130527 0.7095467
## diff_p 0.1415261 0.01395193 0.1192970 0.1636831
```

Posterior Validation Check



- The model makes terrible predictions
- Only 2 departments admit a lower percentage of women than men

- Our model didn't look department by department.
  - It looked at all women applicants vs. all men, without considering that more women apply to some departments and more men apply to others:



  - 51% of men apply to A or B
  - 54% of women apply to C or E

A Better Model

```
d_grad |>
  group_by(dept) |>
  mutate(pg = applications / sum(applications)) |>
  ungroup() |>
  select(dept, applicant.gender, pg) |>
  pivot_wider(names_from = "dept", values_from = "pg") |>
  kable(digits = 2)
```

applicant.gender	A	B	C	D	E	F
male	0.88	0.96	0.35	0.53	0.33	0.52
female	0.12	0.04	0.65	0.47	0.67	0.48

```
dat_list <- d_grad |>
  mutate(dept = ordered(dept), dept_id = as.integer(dept)) |>
  select(admit, applications, gid, dept_id)

mdl_gs_2 <- ulam(
  alist(
    admit ~ dbinom(applications, p),
    logit(p) <- a[gid] + delta[dept_id],
    a[gid] ~ dnorm(0, 1.5),
    delta[dept_id] ~ dnorm(0, 1.5)
  ), data = dat_list, chains = 4, cores=4, iter=4000)

#
```

```
precis_show(precis(mdl_gs_2, depth=2, digits=2))
```

```
##            mean   sd  5.5% 94.5% rhat ess_bulk
## a[1]      -0.51 0.53 -1.35  0.31 1.01   600.24
## a[2]      -0.41 0.53 -1.26  0.41 1.01   594.08
## delta[1]   1.09 0.53  0.27  1.94 1.00   611.67
## delta[2]   1.05 0.53  0.22  1.90 1.00   609.90
## delta[3]  -0.17 0.53 -0.99  0.68 1.01   603.48
## delta[4]  -0.20 0.53 -1.02  0.65 1.01   600.74
## delta[5]  -0.65 0.53 -1.48  0.21 1.00   603.42
## delta[6]  -2.20 0.54 -3.06 -1.33 1.00   638.64
```

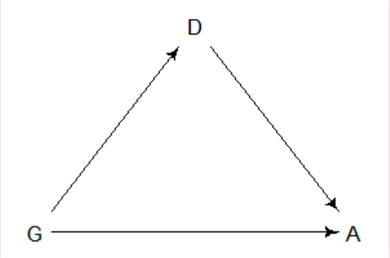
```
post <- tidy_draws(mdl_gs_2)
diff_a <- pull(post, "a[1]") - pull(post, "a[2]")
diff_p <- inv_logit(pull(post, "a[1]")) -
  inv_logit(pull(post, "a[2]"))
precis(list(diff_a = diff_a, diff_p = diff_p))
```

```
##                mean         sd        5.5%      94.5%
histogram
## diff_a -0.09742307 0.08063406 -0.22599692 0.03122233
▁▁▁▁▁▁▁▅▇▇▅▁▁▁▁▁▁
## diff_p -0.02187250 0.01850105 -0.05231097 0.00695312
▁▁▁▅▇▅▁▁▁
```

- Women preferentially apply to more competitive programs than men
- Once we account for this, there's no significant difference between admissions rates for men and women.

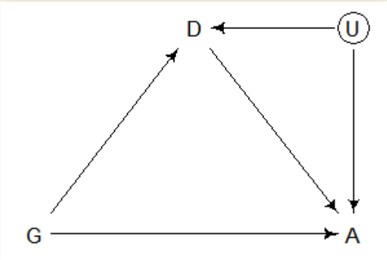
Causal Analysis

- Causal diagram:



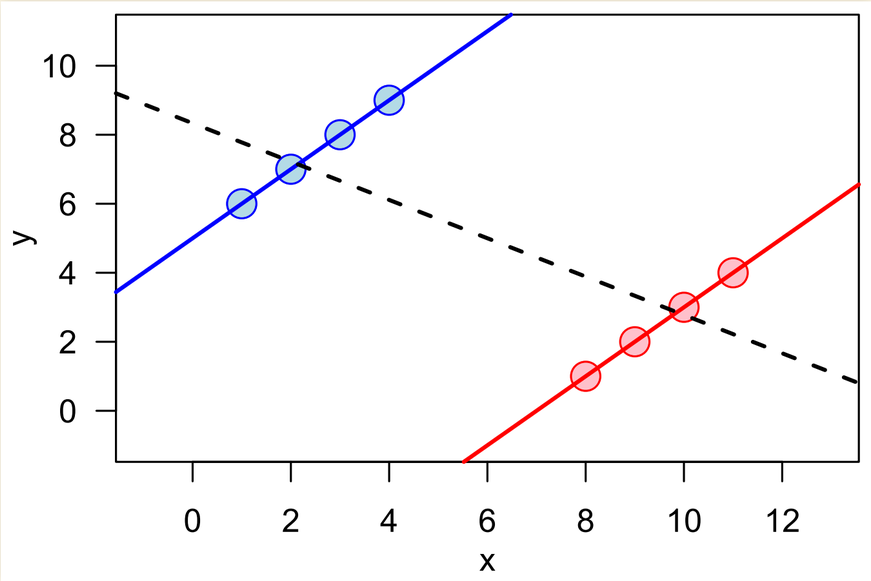
  - We are interested in the direct effect of `G` on `A`
  - But `G` also influences `D`
    - Women and men have different preferences about which graduate programs to apply to
    - And `D` affects `A`:
      - Some programs are more competitive than others

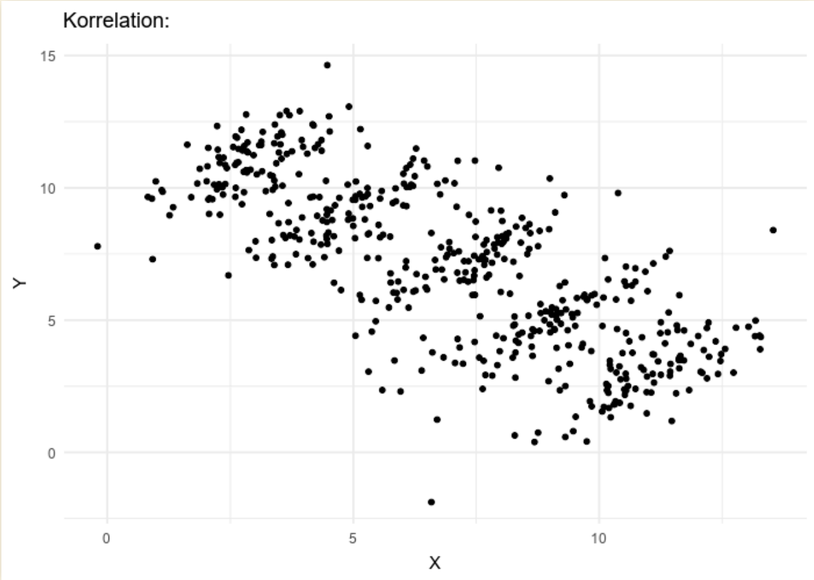
- `G -> D -> A` is a back door, which is open unless we close it by conditioning on `D`
- This analysis does not prove that there was no gender discrimination at Berkeley
  - It shows that we can't prove gender discrimination from the data available
- A more sophisticated analysis could consider whether there are other variables that also affect which departments people apply to



Simpson’s Paradox

- Split data into multiple groups (departments)
  - Each group has one kind of effect.
- When you look at all the data together, it has the opposite effect.
  - If groupings correlate with other variables, ignoring them opens a backdoor
    - Open backdoor causes misleading results





Images: Wikipedia article on Simpson’s Paradox

Other Likelihood Functions

Poisson Distribution

- Poisson
  - In a binomial distribution $\mathsf{Binomial}(N, p)$,
    - Average count = $N \times p$
    - Maximum possible = N
  - Suppose we let N get really large and p get small, while holding the average $N \times p$ constant
    - Now, the maximum becomes very large
    - Probability of N events is p^N, which is very small
    - We can ignore N and focus on the average $N \times P$
- In this limit, the binomial becomes a Poisson distribution $\mathsf{Poisson}(\lambda)$, where $\lambda = N \times p$ is the average.
- An example is radioactive decay.
  - $^{230}\mathsf{Th}$ has a half-life of 75,000 years, or 1.72×10^{12} seconds.
  - An individual nucleus has a probability of 2.93×10^{-13} of decaying in the next second
  - A rock containing one nanomole of $^{230}\mathsf{Th}$ will have an average of 1,760 decays per second.
  - The distribution of decays-per-second will follow $\mathsf{Poisson}(1760.)$

Overdispersed Data

- Sometimes a binomial or Poisson distribution doesn't fit your data as well as you expect.
  - One cause is *rate heterogeneity*
    - Poisson and binomial distributions assume that every event has identical probability
      - Every radionuclide has identical probability of decaying
      - Every coin toss has identical probability of landing "heads"
    - Sometimes events have varying probability:
      - Different individuals within a species can have different probability of dying, even after conditioning on age, sex, etc.
      - Infrastructure failure: Not every power line has the same change of failing in strong winds or ice loading.
  - *Rate heterogeneity* leads to *overdispersed* data:
    - The variance of $\mathsf{Poisson}(\lambda)$ is λ.
    - The variance of $\mathsf{Binomial}(N, p)$ is $N \times p \times (1 - p)$
  - If the variance of the residuals of your model (observations – predictions) is significantly greater than the predicted variance, your data is *overdispersed*
    - Consider a likelihood function that accounts for *rate heterogeneity*

Overdispersed Binomial Distributions

- Negative binomial
  - Also called gamma-Poisson
  - A mixture of Poisson distributions with different rates.
  - More on mixture models on Wednesday
  - Negative binomial distribution is also useful because it represents the distribution of how many trials you expect before a certain number of events happen.
    - How many times do you have to toss a coin before you get a total of six heads?
- Beta binomial
  - A mixture of binomial distributions with the same N, but p is drawn randomly from a β distribution.
    - Example: J.M. Gilligan *et al.*, "Urban Water Conservation Policies in the United States", *Earth's Future* **6**, 955–967 2018. doi: 10.1029/2017EF000797
    - Analyzed the 195 largest cities in 45 states, counting the number of water conservation policies they adopted (out of 53 possible).
    - $\text{Binomial}(53, p)$
    - But the data were overdispersed: Not all policies were equally likely
    - A *beta-binomial* distribution fit the data much better.

Multinomial Models

- Bernoulli, binomial, Poisson, and their variants consider binary outcomes:
  - heads vs. tails, land vs. water, dead vs. alive, left vs. right lever, admit vs. reject, a nucleus decays or it doesn't, etc.
- Sometimes you want to predict outcomes that have more categories:
  - A die can come up 1 through 6.
  - Predict how many marbles of each color you will draw from a bag containing many colors of marbles.
  - Predict prevalence of different minerals in certain rocks.
  - Predict which species will be observed in different habitats.